

Test 3

Aditya

September 2026

1 Neural Network

the input is $x_1=0$ and $x_2=3$ with weights $w_1=0.1$ and $w_2=0.2$ and bias is $b=0.5$

$$z = 0(1) + 0.2(3) + 0.5$$

$$z = 1.1$$

the sigmoid function would be

$$\text{sigma}(z) = 1/(1 + e^{-z})$$

$$z = 0.7503$$

we need a activation function for the neuron to compute the right output. if not the neuron would just produce weighted sum.

2 inference

real-world inference systems try to batch because batching reduces the risk of the system from falling or crashing. in batching the model introduces more requests by queueing them while the other requests are being processed. this helps the model respond faster